Project 2

1 Data Description

This project uses the DAVIS and KIBA drug–target binding datasets to study affinity patterns and build a simple collaborative filtering recommender. Both datasets are publicly available research benchmarks. Each observation links a small-molecule ligand to a protein target and a binding score.

Data links:

Source 1: Kaggle.

https://www.kaggle.com/datasets/rajaryan2315/davis-and-kiba-datasets

Source 2: Off-the-shelf from previous research, can use directly.

Dataset 1: DAVIS

https://github.com/hkmztrk/DeepDTA/tree/master/data/davis

Dataset 2: KIBA

https://github.com/hkmztrk/DeepDTA/tree/master/data/kiba

DAVIS Dataset Structure

- **Ligands and targets:** 68 ligands and 442 protein targets.
- **Observations:** 30,056 ligand–target pairs; the supplied matrix is complete.
- **Scores:** Dissociation constants Kd in nM. Use pKd for analysis and modelling.

KIBA Dataset Structure

- **Ligands and targets:** 2,111 ligands and 229 protein targets.
- **Observations:** 118,254 measured pairs in a sparse matrix. Missing entries are not zero scores.
- **Scores:** Composite KIBA activity scores in the supplied transformed benchmark scale.

Background Knowledge

- **Drug–target binding:** A ligand is a small molecule that may bind to a target protein. One ligand may interact with several targets.
- **Binding affinity:** Lower Kd indicates stronger binding. Higher pKd indicates stronger binding. E.g., convert in DAVIS using $pKd = -\log10(Kd / 10^9) = 9 - \log10(Kd)$, with Kd in nM.
- **Composite activity score:** KIBA combines information from Ki, Kd and IC50 measurements. In the supplied version, higher scores indicate stronger affinity.

Tips

1. As a shortcut, it is suggested to use simplified molecular input line entry system (SMILES) drug sequences and amino-acid alphabet target sequences as input, and the scores as the labels, for your model(s) along the project, although for the first two tasks, ID-based implementations are also suitable.
2. There are duplicated pairs in the dataset, depending on IDs or sequences you use as inputs. You may need to conduct necessary deduplication and may average the scores of repeated samples.

Reference:

[1] Öztürk, Hakime, Arzucan Özgür, and Elif Ozkirimli. “DeepDTA: deep drug–target binding affinity prediction.” *Bioinformatics* 34.17 (2018): i821–i829.

2 Task Description

2.1 Analysis of Drug–Target Datasets

● Objective:

Explore score distributions and observation of the drug-target pairs. Explain the main features and key differences between DAVIS and KIBA.

● Steps:

1. Download Two Datasets and Overview

Download both datasets into data/ of your project folder. Show necessary **examples in table(s) to reflect the data format.**

2. Data Cleaning

Apply fundamental cleaning methods, e.g., exclude non-finite scores and require positive DAVIS Kd. Keep valid and necessary extreme or zero scores. You also need to conduct necessary deduplication or average the scores of repeated samples. **Report original and deduplicated ligand, target and observed-pair counts in table(s).**

3. Visualization

- **Heatmaps:** Plot a separate ligand–target heatmap for each dataset: DAVIS pKd and supplied KIBA scores. Clearly distinguish missing entries.
- **Distributions:** Plot the observed score distribution for each dataset, preferably using KDE. Label the score scale and density axis.
- **Any analytical tools you learned in this course:** Include helpful methods you think that can support your findings on ligand and target properties. Please refrain from fitting or finetuning large models here because this is a step to help explaining and understanding your dataset.

● Tips:

1. Discuss abnormality from weak-binding or screening-limit records. Never silently remove them.
2. Report KIBA sparsity as missing positions divided by all matrix positions can be helpful. **Missing observations do not establish non-binding.**

In the report, document your cleaning, present figures and tables, and explain what they show.

Code Reuse: Suggested to use matplotlib | seaborn | plotly for visualization and pandas for tables in a Jupyter notebook. Write your own code and analysis, and cite sources of key insights or reused code above the relevant implementation in your report.

2.2 Recommending a Drug to a Given Target

● Objective:

For a ligand represented in training, predict scores and rank targets whose scores are held out. Assess the value of collaborative filtering beyond looking up known measurements.

● Method:

Implement a collaborative filtering to address unseen drug-target pairs. Discuss it on DAVIS and KIBA.

● Steps:

1. Data Preprocessing and Data Split

Who are "items", and who are "users" in this drug-target case? Use DAVIS pKd and the supplied KIBA scores. Reuse the fixed training, validation and test partitions (e.g., 70% | 15% | 15%, control random seed strictly). No ligand–target pair may occur in multiple partitions. All evaluation ligands and targets must occur in training. Fit model parameters on training observations only. Select hyperparameters on validation data, then evaluate once on test data.

2. Metrics

- **MSE:** Report mean squared error between predicted and true scores for training, validation and test. Lower values are better.
- **CI:** Report the Concordance Index, i.e., ordering agreement between pairs with different true scores. *Tips: Kendall τ could be used to compute $CI = \left(\tau \times \sqrt{B/A} + 1\right)/2$, where A and B are respectively the number of pairs with different true scores and different predicted scores.*
- **Recommendation quality:** Hit@1, Hit@5 of recommendation, and other metrics you like.

3. Save the Models

Save separate checkpoints as saved_models/davis_cf.npz and saved_models/kiba_cf.npz for instance. Preserve learned CF state and score-scale information. Verify that reloading reproduces predictions.

4. Case Study

At the end of Task 2, analyse three fixed test ligands per dataset. Generate all case predictions by loading the corresponding saved model from disk.

E.g., if your test set is not including:

- **DAVIS:** 10127622, 11338033, 44259.
- **KIBA:** CHEMBL1967538, CHEMBL2004544, CHEMBL565612.

You could use them in this case study and **describe the reason for choosing these cases**. For each ligand, show recommendations, predicted and true scores, MSE/CI and ranking metrics, including the predicted rank of the true best target. Label this target "correct" and other candidates "wrong".

● Tips:

1. The recommended results are not necessarily correct but must address various conditions.
2. Is the recommender necessary here? Either a positive or a negative conclusion is acceptable if supported by your own **theoretical analysis** or extended experiments, rather than an opinion alone.
3. Do address both datasets. KIBA is not only larger but also sparse. Explain how missing observations and abnormal values affect similarity, available neighbours and fallback predictions.
4. For each dataset, explain why your method is not simply a query of known answers. Use held-out observations and a baseline to support your conclusion.

In the report, explain your method, report the results and discuss the cases and limitations.

2.3 Open Advanced Task: Cold-Target Generalization

● Objective

Have you noticed that your old implementation of CF may be unable to generalize to unseen target recommendations? What if a target in the test set has no affinity measurements in the training set? CF relies on neighbors' known scores for the target. If a new target has no recorded scores, even if its ID is included in the model, the prediction falls back to the ligand's mean training score and cannot distinguish between different new targets.

To address this limitation, please try to provide a solution based on the given DAVIS-KIBA dataset. The solution includes but not limited to traditional analytics, machine learning, deep learning, and LLM-related pretraining and finetuning tasks. Implement your solution and provide some experimental results to show your solution works.

Reference:

[1] Kim, Youngoh, et al. "MixingDTA: improved drug–target affinity prediction by extending mixup with guilt-by-association." *Bioinformatics* 41.Supplement_1 (2025): i105-i114.

In the report, you need to explain the algorithm you design. You also need to document the results of your algorithm.

Weight of grading:

Task 1: 30%

Task 2: 40%

Task 3: 30%

These gradings are distributed according to tasks. So make sure the code, the written report, and the presentation (by video) fully convey your approaches and findings.

What to deliver:

Code: for the three tasks.

Report: The final report is up to 8 A4 pages excluding acknowledgements and reference (not necessary to write 8 pages. The page limit does not include front page. Appendix outside these pages will not contribute to your grade.)

*** If there is unequal contribution,** include a table showing the percentage contribution and justification, supported by the email confirming all group members agree with the distribution. In the end of report, please include the individual contribution claims in the following format:

Member Name	Contribution (%)	Justification
Member name 1	p%	list of contributions to the project.

Up to 5 minutes presentation by video: In the short video, you can capture screen to do a demo of your code to show it works. Or you can do a highlight presentation by slides and record it if you like.

Project are done in groups (the same group list as in project 1). Discussions with other students are allowed, but each group has to write your own code.

Submission Instructions

Deadline: 16 Nov. 0:00 AM Singapore Time.

Deliverables: Code + Report + Video (Link).

Code & Report: Submit softcopy through NTUlearn.

Video: Please do not upload the video directly to NTUlearn. Instead, each group should upload their video to their own cloud storage (e.g., OneDrive, Google Drive, or other equivalent service) and generate a shareable link. The link should then be submitted via NTUlearn together with your report and code.

The TA will access your video online using the shared link. Please make sure the link is valid until grading is completed.

Grading will consider report + code + video (5 mins)

NOTE:

1. **MOSS:** Sharing code with your classmates is not acceptable! All programs will be screened using the Moss (Measure of Software Similarity.) system.
2. **You are not allowed to share your project code on the web publicly.**

TA for projects:

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If you have questions, please email above. TAs can only provide some consultation for projects, but you should NOT expect TAs to help to do any part of your project.